

CW Approximation

A map $f: X \rightarrow Y$ is called a **weak homotopy equivalence** if it induces isomorphisms $\pi_n(X, x_0) \rightarrow \pi_n(Y, f(x_0))$ for all $n \geq 0$ and all choices of basepoint x_0 . Whitehead's theorem can be restated as saying that a weak homotopy equivalence between CW complexes is a homotopy equivalence. It follows easily that this holds also for spaces homotopy equivalent to CW complexes. In general, however, weak homotopy equivalence is strictly weaker than homotopy equivalence. For example, there exist noncontractible spaces whose homotopy groups are all trivial, such as the 'quasi-circle' according to an exercise at the end of this section, and for such spaces a map to a point is a weak homotopy equivalence that is not a homotopy equivalence.

One of the more important results in this subsection is that for every space X there is a CW complex Z and a weak homotopy equivalence $f: Z \rightarrow X$. Such a map $f: Z \rightarrow X$ is called a **CW approximation** to X . A weak homotopy equivalence induces isomorphisms on all homology and cohomology groups, as we will show, so CW approximations allow many general statements in algebraic topology to be reduced to the case of CW complexes, where one can often make cell-by-cell arguments.

The technique for constructing CW approximations can be used to do other things as well. For a start, one could try for a relative version in which X is assumed to contain a subspace A which is already a CW complex and Z is constructed to contain A as a subcomplex, with $f: Z \rightarrow X$ restricting to the identity map on A . Next, if (X, A) is n -connected, one could try to make (Z, A) n -connected in the strong geometric sense that all cells of $Z - A$ have dimension greater than n . In fact, it turns out to be possible to do a construction satisfying the latter condition even if (X, A) is not n -connected, but in this case $\pi_i(Z)$ for $i < n$ will be isomorphic to $\pi_i(A)$ rather than to $\pi_i(X)$.

Here is a definition that is sufficiently general to cover all these situations. Given a pair (X, A) where the subspace $A \subset X$ is a nonempty CW complex, an **n -connected CW model for (X, A)** is an n -connected CW pair (Z, A) and a map $f: Z \rightarrow X$ with $f|_A$ the identity, such that $f_*: \pi_i(Z) \rightarrow \pi_i(X)$ is an isomorphism for $i > n$ and an injection for $i = n$, for all choices of basepoint. Since (Z, A) is n -connected, the map $\pi_i(A) \rightarrow \pi_i(Z)$ is an isomorphism for $i < n$ and a surjection for $i = n$. In the critical dimension n , the maps $A \hookrightarrow Z \xrightarrow{f} X$ induce a composition $\pi_n(A) \rightarrow \pi_n(Z) \rightarrow \pi_n(X)$ factoring the map $\pi_n(A) \rightarrow \pi_n(X)$ as a surjection followed by an injection, just as any homomorphism $\varphi: G \rightarrow H$ can be factored (uniquely) as a surjection $\varphi: G \rightarrow \text{Im } \varphi$ followed by an injection $\text{Im } \varphi \hookrightarrow H$. One can think of Z as a sort of homotopy-theoretic hybrid of A and X . As n increases, the hybrid looks more and more like A , and less and less like X .

This definition specializes to the earlier notion of a CW approximation by taking $n = 0$ and letting A consist of one point in each path-component of X . This forces $f_*: \pi_0(Z) \rightarrow \pi_0(X)$ to be surjective as well as injective.

Proposition 4.13. *For every pair (X, A) with A a nonempty CW complex there exist n -connected CW models $f: (Z, A) \rightarrow (X, A)$ for all $n \geq 0$, and these models can be chosen to have the additional property that Z is obtained from A by attaching cells of dimension greater than n .*

Note that the condition that $Z - A$ consists of cells of dimension greater than n automatically implies that (Z, A) is n -connected, by cellular approximation.

Proof: We will construct Z as a union of subcomplexes $A = Z_n \subset Z_{n+1} \subset \cdots$ with Z_k obtained from Z_{k-1} by attaching k -cells. Suppose inductively that we have already constructed Z_k and a map $f: Z_k \rightarrow X$ restricting to the identity on A and such that the induced map on π_i is an injection for $n \leq i < k$ and a surjection for $n < i \leq k$, with respect to a choice of basepoint 0-cell x_y in each component A_y of A . The induction begins with $k = n$ and $Z_n = A$, when these conditions are vacuous.

For the induction step, choose cellular maps $\varphi_\alpha: S^k \rightarrow Z_k$ representing generators for the kernel of $f_*: \pi_k(Z_k, x_y) \rightarrow \pi_k(X, x_y)$, for all y . Attach cells e_α^{k+1} to Z_k via these maps φ_α , and call the resulting complex Y_{k+1} . Since the compositions $f\varphi_\alpha$ are nullhomotopic, we can extend f over Y_{k+1} . The map $f_*: \pi_k(Y_{k+1}, x_y) \rightarrow \pi_k(X, x_y)$ is then injective since each element of the kernel is represented by a cellular map, with image in Z_k , and such maps are nullhomotopic in Y_{k+1} by construction. The extended f still induces a surjection on π_k since the composition $\pi_k(Z_k) \rightarrow \pi_k(Y_{k+1}) \rightarrow \pi_k(X)$ is surjective. The homotopy groups π_i for $i < k$ are not affected by attaching the cells e_α^{k+1} . When $k = 0$ the construction needs to be done differently since π_0 has no group structure. Instead, we form Y_1 by attaching 1-cells joining all basepoint 0-cells x_y lying in the same path-component of X .

Next, choose maps $\psi_\beta: S^{k+1} \rightarrow X$ generating $\pi_{k+1}(X, x_y)$ for all y . Let Z_{k+1} be the wedge sum of Y_{k+1} with spheres S_β^{k+1} at the appropriate basepoints x_y , and extend f over Z_{k+1} by letting it equal ψ_β on S_β^{k+1} . This guarantees that the induced map $f_*: \pi_{k+1}(Z_{k+1}, x_y) \rightarrow \pi_{k+1}(X, x_y)$ is surjective. The inclusion $Y_{k+1} \hookrightarrow Z_{k+1}$ induces an isomorphism on π_i for $i \leq k$, surjectivity coming from cellular approximation and injectivity from a retraction of Z_{k+1} onto Y_{k+1} . This finishes the induction step.

Since the maps $f_*: \pi_i(Z, x_y) \rightarrow \pi_i(X, x_y)$ depend only on the $(i+1)$ -skeleton of Z , they are isomorphisms for all $i > n$ and injective for $i = n$. This holds in fact for all basepoints in Z , not just the x_y 's, since every point in Z is joined by a path to some x_y . $\square$

Example 4.14. When X is path-connected and A is a point, the construction of a 0-connected CW model for (X, A) gives a CW approximation to X with a single 0-cell and all higher cells attached by basepoint-preserving maps. In particular, any connected CW complex is homotopy equivalent to a CW complex with these properties.

Example 4.15. One can also apply the proposition to obtain a CW approximation to an arbitrary pair (X, X_0) . First construct a CW approximation $f_0: Z_0 \rightarrow X_0$, then form

a 0-connected CW model $(Z, Z_0) \rightarrow (M, Z_0)$ where M is the mapping cylinder of the composition of f_0 with the inclusion $X_0 \hookrightarrow X$. Composing the map $Z \rightarrow M$ with the retraction of M onto X , we obtain an extension of f_0 to a CW approximation $f: Z \rightarrow X$. It follows from the five-lemma that the map $(Z, Z_0) \rightarrow (X, X_0)$ induces isomorphisms on relative as well as absolute homotopy groups.

Here is a rather different application of the preceding proposition, giving a more geometric interpretation to n -connectedness:

Corollary 4.16. *If (X, A) is an n -connected CW pair, then there exists a CW pair $(Z, A) \simeq (X, A) \text{ rel } A$ such that all cells of $Z - A$ have dimension greater than n .*

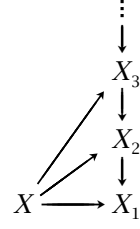
Proof: An n -connected CW approximation $f: (Z, A) \rightarrow (X, A)$ given by the preceding proposition will do the trick. First we check that f induces isomorphisms $\pi_i(Z) \approx \pi_i(X)$ for all i . This is true for $i > n$ by definition, and for $i < n$ it holds since both inclusions $A \hookrightarrow Z$ and $A \hookrightarrow X$ induce isomorphisms on these lower homotopy groups. For $i = n$, f induces an injection on π_n by definition, and since the inclusion $A \hookrightarrow X$ induces a surjection on π_n , so does f via the composition $\pi_n(A) \rightarrow \pi_n(Z) \rightarrow \pi_n(X)$.

Since f induces isomorphisms on all homotopy groups, it is a homotopy equivalence. To see that it is a homotopy equivalence $\text{rel } A$, form a quotient space W of the mapping cylinder M_f by collapsing each segment $\{a\} \times I$ to a point, for $a \in A$. Assuming f has been made cellular, W is a CW complex containing X and Z as subcomplexes, and W deformation retracts to X just as M_f does. Also, $\pi_i(W, Z) = 0$ for all i since f induces isomorphisms on all homotopy groups, so W deformation retracts onto Z . These two deformation retractions of W onto X and Z are stationary on A , hence give a homotopy equivalence $X \simeq Z \text{ rel } A$. $\square$

Example 4.17: Postnikov Towers. For a CW complex X , which we may as well take to be connected, let us construct a sequence of spaces X_n such that $\pi_i(X_n) \approx \pi_i(X)$ for $i \leq n$ and $\pi_i(X_n) = 0$ for $i > n$. Choose cellular maps $\varphi_\alpha: S^{n+1} \rightarrow X$ generating π_{n+1} and use these to attach cells e_α^{n+2} to X , forming a CW complex Y . By cellular approximation the inclusion $X \hookrightarrow Y$ induces isomorphisms on π_i for $i \leq n$, and $\pi_{n+1}(Y) = 0$ since any element of $\pi_{n+1}(Y)$ is represented by a map to X by cellular approximation, and such maps are nullhomotopic in Y by construction. Now the process can be repeated with Y in place of X and n replaced by $n + 1$ to make a space with π_{n+2} zero as well as π_{n+1} , by attaching $(n + 3)$ -cells. After infinitely many iterations we have enlarged X to a CW complex X_n such that the inclusion $X \hookrightarrow X_n$ induces an isomorphism on π_i for $i \leq n$ and $\pi_i(X_n) = 0$ for $i > n$. This construction is in fact a special case of the construction of CW models, with (X_n, X) an $(n + 1)$ -connected CW model for (CX, X) with CX the cone on X .

The inclusion $X \hookrightarrow X_n$ extends to a map $X_{n+1} \rightarrow X_n$ since X_{n+1} is obtained from X by attaching cells of dimension $n + 3$ and greater, and $\pi_i(X_n) = 0$ for $i > n$ so we

can apply Lemma 4.7, the extension lemma. Thus we have a commutative diagram as at the right. This is called a *Postnikov tower* for X . One can regard the spaces X_n as truncations of X which provide successively better approximations to X as n increases. Postnikov towers turn out to be quite powerful tools for proving general theorems, and we will study them further in §4.3.



After this example one may wonder whether n -connected CW models (Z_n, A) for an arbitrary pair (X, A) always fit into a tower. The following proposition will allow us to construct such towers, among other things.

Proposition 4.18. *Suppose we are given:*

- (i) *an n -connected CW model $f: (Z, A) \rightarrow (X, A)$,*
- (ii) *an n' -connected CW model $f': (Z', A') \rightarrow (X', A')$,*
- (iii) *a map $g: (X, A) \rightarrow (X', A')$.*

$$\begin{array}{ccc} Z & \xrightarrow{f} & X \\ h \downarrow & & \downarrow g \\ Z' & \xrightarrow{f'} & X' \end{array}$$

Then if $n \geq n'$, there is a map $h: Z \rightarrow Z'$ such that $h|_A = g$ and $gf \simeq f'h \text{ rel } A$, so the diagram above is commutative up to homotopy $\text{rel } A$. Furthermore, such a map h is unique up to homotopy $\text{rel } A$.

Proof: By Corollary 4.16 we may assume all cells of $Z - A$ have dimension greater than n . Let W be the quotient space of the mapping cylinder of f' obtained by collapsing each line segment $\{a'\} \times I$ to a point, for $a' \in A'$. We can think of W as a relative mapping cylinder, and like the ordinary mapping cylinder, W contains copies of Z' and X' , the latter as a deformation retract. The assumption that (Z', A') is an n' -connected CW model for (X', A') implies that the relative groups $\pi_i(W, Z')$ are zero for $i > n'$.

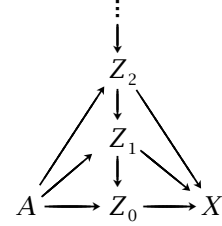
Via the inclusion $X' \hookrightarrow W$ we can view gf as a map $Z \rightarrow W$. As a map of pairs $(Z, A) \rightarrow (W, Z')$, gf is homotopic $\text{rel } A$ to a map h with image in Z' , by the compression lemma and the hypothesis $n \geq n'$. This proves the first assertion. For the second, suppose h_0 and h_1 are two maps $Z \rightarrow Z'$ whose compositions with f' are homotopic to $gf \text{ rel } A$. Thus if we regard h_0 and h_1 as maps to W , they are homotopic $\text{rel } A$. Such a homotopy gives a map $(Z \times I, Z \times \partial I \cup A \times I) \rightarrow (W, Z')$, and by the compression lemma again this map can be deformed $\text{rel } Z \times \partial I \cup A \times I$ to a map with image in Z' , which gives the desired homotopy $h_0 \simeq h_1 \text{ rel } A$. $\square$

Corollary 4.19. *An n -connected CW model for (X, A) is unique up to homotopy equivalence $\text{rel } A$. In particular, CW approximations to spaces are unique up to homotopy equivalence.*

Proof: Given two n -connected CW models (Z, A) and (Z', A) for (X, A) , we apply the proposition twice with g the identity map to obtain maps $h: Z \rightarrow Z'$ and $h': Z' \rightarrow Z$. The uniqueness statement gives homotopies $hh' \simeq \mathbb{1}$ and $h'h \simeq \mathbb{1} \text{ rel } A$. $\square$

Taking $n = n'$ in the proposition, we obtain also a functoriality property for n -connected CW models. For example, a map $X \rightarrow X'$ induces a map of CW approximations $Z \rightarrow Z'$, which is unique up to homotopy.

The proposition allows us to relate n -connected CW models (Z_n, A) for (X, A) for varying n , by means of maps $Z_n \rightarrow Z_{n-1}$ that form a tower as shown in the diagram, with commutative triangles on the left and homotopy-commutative triangles on the right. We can make the triangles on the right strictly commutative by replacing the maps $Z_n \rightarrow X$ by the compositions through Z_0 .



Example 4.20: Whitehead Towers . If we take X to be an arbitrary CW complex with the subspace A a point, then the resulting tower of n -connected CW models amounts to a sequence of maps

$$\cdots \rightarrow Z_2 \rightarrow Z_1 \rightarrow Z_0 \rightarrow X$$

with Z_n n -connected and the map $Z_n \rightarrow X$ inducing an isomorphism on all homotopy groups π_i with $i > n$. The space Z_0 is path-connected and homotopy equivalent to the component of X containing A , so one may as well assume Z_0 equals this component. The next space Z_1 is simply-connected, and the map $Z_1 \rightarrow X$ has the homotopy properties of the universal cover of the component Z_0 of X . For larger values of n one can by analogy view the map $Z_n \rightarrow X$ as an ' n -connected cover' of X . For $n > 1$ these do not seem to arise so frequently in nature as in the case $n = 1$. A rare exception is the Hopf map $S^3 \rightarrow S^2$ defined in Example 4.45, which is a 2-connected cover.

Now let us show that CW approximations behave well with respect to homology and cohomology:

Proposition 4.21. *A weak homotopy equivalence $f: X \rightarrow Y$ induces isomorphisms $f_*: H_n(X; G) \rightarrow H_n(Y; G)$ and $f^*: H^n(Y; G) \rightarrow H^n(X; G)$ for all n and all coefficient groups G .*

Proof: Replacing Y by the mapping cylinder M_f and looking at the long exact sequences of homotopy, homology, and cohomology groups for (M_f, X) , we see that it suffices to show:

- If (Z, X) is an n -connected pair of path-connected spaces, then $H_i(Z, X; G) = 0$ and $H^i(Z, X; G) = 0$ for all $i \leq n$ and all G .

Let $\alpha = \sum_j n_j \sigma_j$ be a relative cycle representing an element of $H_k(Z, X; G)$, for singular k -simplices $\sigma_j: \Delta^k \rightarrow Z$. Build a finite Δ -complex K from a disjoint union of k -simplices, one for each σ_j , by identifying all $(k-1)$ -dimensional faces of these k -simplices for which the corresponding restrictions of the σ_j 's are equal. Thus the σ_j 's induce a map $\sigma: K \rightarrow Z$. Since α is a relative cycle, $\partial\alpha$ is a chain in X . Let

$L \subset K$ be the subcomplex consisting of $(k-1)$ -simplices corresponding to the singular $(k-1)$ -simplices in $\partial\alpha$, so $\sigma(L) \subset X$. The chain α is the image under the chain map $\sigma_\#$ of a chain $\tilde{\alpha}$ in K , with $\partial\tilde{\alpha}$ a chain in L . In relative homology we then have $\sigma_*[\tilde{\alpha}] = [\alpha]$. If we assume $\pi_i(Z, X) = 0$ for $i \leq k$, then $\sigma: (K, L) \rightarrow (Z, X)$ is homotopic rel L to a map with image in X , by the compression lemma. Hence $\sigma_*[\tilde{\alpha}]$ is in the image of the map $H_k(X, X; G) \rightarrow H_k(Z, X; G)$, and since $H_k(X, X; G) = 0$ we conclude that $[\alpha] = \sigma_*[\tilde{\alpha}] = 0$. This proves the result for homology, and the result for cohomology then follows by the universal coefficient theorem. $\square$

CW approximations can be used to reduce many statements about general spaces to the special case of CW complexes. For example, the cup product version of the Künneth formula in Theorem 3.16, asserting that $H^*(X \times Y; R) \approx H^*(X; R) \otimes H^*(Y; R)$ under certain conditions, can now be extended to non-CW spaces since if X and Y are CW approximations to spaces Z and W , respectively, then $X \times Y$ is a CW approximation to $Z \times W$. Here we are giving $X \times Y$ the CW topology rather than the product topology, but this has no effect on homotopy groups since the two topologies have the same compact sets, as explained in the Appendix. Similarly, the general Künneth formula for homology in §3.B holds for arbitrary products $X \times Y$.

The condition for a map $Y \rightarrow Z$ to be a weak homotopy equivalence involves only maps of spheres into Y and Z , but in fact weak homotopy equivalences $Y \rightarrow Z$ behave nicely with respect to maps of arbitrary CW complexes into Y and Z , not just spheres. The following proposition gives a precise statement, using the notations $[X, Y]$ for the set of homotopy classes of maps $X \rightarrow Y$ and $\langle X, Y \rangle$ for the set of basepoint-preserving-homotopy classes of basepoint-preserving maps $X \rightarrow Y$. (The notation $\langle X, Y \rangle$ is not standard, but is intended to suggest ‘pointed homotopy classes.’)

Proposition 4.22. *A weak homotopy equivalence $f: Y \rightarrow Z$ induces bijections $[X, Y] \rightarrow [X, Z]$ and $\langle X, Y \rangle \rightarrow \langle X, Z \rangle$ for all CW complexes X .*

Proof: Consider first $[X, Y] \rightarrow [X, Z]$. We may assume f is an inclusion by replacing Z by the mapping cylinder M_f as usual. The groups $\pi_n(Z, Y, y_0)$ are then zero for all n and all basepoints $y_0 \in Y$, so the compression lemma implies that any map $X \rightarrow Z$ can be homotoped to have image in Y . This gives surjectivity of $[X, Y] \rightarrow [X, Z]$. A relative version of this argument shows injectivity since we can deform a homotopy $(X \times I, X \times \partial I) \rightarrow (Z, Y)$ to have image in Y .

In the case of $\langle X, Y \rangle \rightarrow \langle X, Z \rangle$ the same argument applies if M_f is replaced by the reduced mapping cylinder, the quotient of M_f obtained by collapsing the segment $\{y_0\} \times I$ to a point, for y_0 the basepoint of Y . This collapsed segment then serves as the common basepoint of Y , Z , and the reduced mapping cylinder. The reduced mapping cylinder deformation retracts to Z just as the unreduced one does, but with the advantage that the basepoint does not move. $\square$

Exercises

1. Suppose a sum $f +' g$ of maps $f, g: (I^n, \partial I^n) \rightarrow (X, x_0)$ is defined using a coordinate of I^n other than the first coordinate as in the usual sum $f + g$. Verify the formula $(f + g) +' (h + k) = (f +' h) + (g +' k)$, and deduce that $f +' k \simeq f + k$ so the two sums agree on $\pi_n(X, x_0)$, and also that $g +' h \simeq h + g$ so the addition is abelian.
2. Show that if $\varphi: X \rightarrow Y$ is a homotopy equivalence, then the induced homomorphisms $\varphi_*: \pi_n(X, x_0) \rightarrow \pi_n(Y, \varphi(x_0))$ are isomorphisms for all n . [The case $n = 1$ is Proposition 1.18.]
3. For an H-space (X, x_0) with multiplication $\mu: X \times X \rightarrow X$, show that the group operation in $\pi_n(X, x_0)$ can also be defined by the rule $(f + g)(x) = \mu(f(x), g(x))$.
4. Let $p: \tilde{X} \rightarrow X$ be the universal cover of a path-connected space X . Show that under the isomorphism $\pi_n(X) \approx \pi_n(\tilde{X})$, which holds for $n \geq 2$, the action of $\pi_1(X)$ on $\pi_n(X)$ corresponds to the action of $\pi_1(X)$ on $\pi_n(\tilde{X})$ induced by the action of $\pi_1(X)$ on $\tilde{X}$ as deck transformations. More precisely, prove a formula like $y p_*(\alpha) = p_*(\beta_{\tilde{y}}(y_*(\alpha)))$ where $y \in \pi_1(X, x_0)$, $\alpha \in \pi_n(\tilde{X}, \tilde{x}_0)$, and y_* denotes the homomorphism induced by the action of y on $\tilde{X}$.
5. For a pair (X, A) of path-connected spaces, show that $\pi_1(X, A, x_0)$ can be identified in a natural way with the set of cosets αH of the subgroup $H \subset \pi_1(X, x_0)$ represented by loops in A at x_0 .
6. If $p: (\tilde{X}, \tilde{A}, \tilde{x}_0) \rightarrow (X, A, x_0)$ is a covering space with $\tilde{A} = p^{-1}(A)$, show that the map $p_*: \pi_n(\tilde{X}, \tilde{A}, \tilde{x}_0) \rightarrow \pi_n(X, A, x_0)$ is an isomorphism for all $n > 1$.
7. Extend the results proved near the beginning of this section for the change-of-basepoint maps β_y to the case of relative homotopy groups.
8. Show the sequence $\pi_1(X, x_0) \rightarrow \pi_1(X, A, x_0) \xrightarrow{\partial} \pi_0(A, x_0) \rightarrow \pi_0(X, x_0)$ is exact.
9. Suppose we define $\pi_0(X, A, x_0)$ to be the quotient set $\pi_0(X, x_0)/\pi_0(A, x_0)$, so that the long exact sequence of homotopy groups for the pair (X, A) extends to $\cdots \rightarrow \pi_0(X, x_0) \rightarrow \pi_0(X, A, x_0) \rightarrow 0$.
 - (a) Show that with this extension, the five-lemma holds for the map of long exact sequences induced by a map $(X, A, x_0) \rightarrow (Y, B, y_0)$, in the following form: One of the maps between the two sequences is a bijection if the four surrounding maps are bijections for all choices of x_0 .
 - (b) Show that the long exact sequence of a triple (X, A, B, x_0) can be extended only to the term $\pi_0(A, B, x_0)$ in general, and that the five-lemma holds for this extension.
10. Show the 'quasi-circle' described in Exercise 7 in §1.3 has trivial homotopy groups but is not contractible, hence does not have the homotopy type of a CW complex.
11. Show that a CW complex is contractible if it is the union of an increasing sequence of subcomplexes $X_1 \subset X_2 \subset \cdots$ such that each inclusion $X_i \hookrightarrow X_{i+1}$ is nullhomotopic, a condition sometimes expressed by saying X_i is contractible in X_{i+1} . An example is